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# DANIEL A FARONBI

danielfaronbi@gmail.com

## SUMMARY

I'm a Music Technology Researcher, Jazz Pianist, and Music Producer. My research involves using signal processing and machine learning techniques for the generation and analysis of music and other audio signals. This includes research in areas like Generative Audio Models, Music Information Retrieval, Computational Musicology, and more.

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## WEBSITES, PORTFOLIOS, PROFILES

- https://danielfaronbi.com/
- https://www.linkedin.com/in/daniel-faronbi-7aa61b184/
- https://github.com/dafaronbi

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## SKILLS

- Python, C++,C, Matlab, Java, Java Script, Kotlin
- Machine Learning/Deep Learning
- Generative Audio Models
- Computational Musicology
- Digital Signal Processing,
- Computer Vision, Music Information Retrieval
- Embedded system development: ARM, AVR, and PIC microcontrollers.
- Linux and Minix Kernel Knowledge
- Front and Back end web development (java script, node js, django)
- PCB design with Eagle Cad and OrCad
- Serial communication protocols: SPI, I2C, UART
- Electronic Circuit Design: amplifiers, voltage regulators, current sources
- Electrical skills with Logic Analyzer usage, Multisim, Oscilloscope usage, Soldering
- Production in Ableton Live, FL Studio, Max MSP, and Pro Tools

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## INDUSTRY EXPERIENCE

**ATG Intern** / Dolby Laboratories - San Fransico, CA *06/2023 - 09/2023*

Audio research as part of the Advanced Technology Group (ATG). Worked on applications for streaming Dolby Atmos audio for collaboratively mixing audio.

**Machine Learning Engineering Intern** / Bose Corporation - Farmingham, MA *06/2022 - 08/2022*

Worked on projects including environmental sound based compositions, source separation for speech, music, and sound effects, and evaluation techniques for deep RMS noise estimators.

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## TEACHING EXPERIENCE

**Music Technology Adjunct Faculty** Fall 2024-Now

(MPATE-GE 2639) Adv Topics Music Tech: C++Audio Application Development:

- This is an advanced graduate-level course covering the C++ programming language, with an emphasis on developing applications for music and audio. Students are expected to have working knowledge of C; this course extends that knowledge to C++, exploring object-oriented topics such as classes, inheritance, function and operator overloading, polymorphism and encapsulation. Students use existing frameworks and libraries to create their own audio applications and audio plug-ins.

(MPATE-GE 2617) C Programming for Music Technology Lab:

- This course provides students with hands-on instruction in algorithmic design & general programming concepts. Example code is implemented, discussed, & demonstrated in detail in order to provide students with step-by-step examples of programming design paradigms

**Sound + Science Student Mentor** Spring 2024-Now
Daniel is the lead mentor for Sound+Science,  a Simons Foundation funded innovative after-school program, research lab, and community in NYC for high school students interested in exploring the unique and deep connection between music, science, and technology. Through a unique curriculum, participants will explore modules like 'The Hidden Codes in Electronic Music', 'From Hip-Hop to AI', and Music Improvisation and the Evolution of the Mathematics of Computation Sound+Science, revealing the profound connections between music, technology, physics and mathematics.
https://soundplusscience.com/

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## LEADERSHIP AND SPEAKING

**NYU GenAudio & AI Club**
Daniel is the president of the The NYU GenAudio & AI club, a group of students and professors who are interested in a variety of applications of generative AI music models. Members of the group read cutting edge papers on generative AI, hear talks from industry and academic experts on AI, and work on relevant group projects focused on a variety of subtasks. The club has organized talks with top generative AI companies like Udio, Sony, Open Art, and Moises as well as top audio researchers like Roger Dannenberg and Meinard Müller. The club also hosts "GenJams" events where participants spend a few hours using generative tools to make a unique piece of art. These events are well attended, ranging from 50-200 attendees per event. These events have had influential host like King Willonius, Time 100 Most Influential in AI; producer of the viral AI track "BBL Drizzy.", and Byron Miller *EVP of Atlantic Music Group*; founder of Miller Tech Talk, bridging entertainment and tech.
**Talks**
Daniel has given a research talks around the world on various topics in AI and music. He has spoken at Stanford University, the Frederick Chhopin University of Music, many Simon's Foundation educational programs, and conferences like ICASPP and AES. His talks have a range of topics. Some are more focused on how jazz harmony relates to natural occurring frequencies in the universe and others are on novel concepts in data science and signal processing for music.
**Music**
Daniel is active jazz pianist, music producer, and composer. He has performed for music groups spanning a wide variety of genres and has received both jazz and classical training. He has worked as a studio manager for Studio 381 at the University of Nebraska and has published two albums. He has shared the stage with jazz heavy hitters such as Benny Golson, Bobby Sanibria, Micah Bell, and Dave Stryker and performed at world famous jazz venues such as The Django and Nublu. He maintains an active performance schedule at various venues in New York City.

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## EDUCATION AND TRAINING

**Ph.D.**: Music Technology
New York University - New York, NY
GPA: 3.422

**Bachelor of Science**: Computer Engineering *05/2021*
University of Nebraska At Lincoln
GPA: 3.658

**Bachelor of Arts**: Music Technology *05/2021*
University of Nebraska At Omaha
Minor: Mathematics

GPA: 3.658

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## RELEVANT COURSES

- 3D Audio
- Audio Recording Techniques I, II, & II | Advanced Computer Music Composition
- Calculus I, II & III
- Communication Systems | Computer Vision
- Data Structures | Deep Learning |Differential Equations
- Digital Design | Digital Synthesis
- Electric Circuits I & III | Electronic Circuits I and II
- Embedded Microcontrollers | Finite Difference Equations
- Fundamentals of Conducting |
- Information Theory | Introduction to Data Science
- Java I and II | Jazz Arranging and Composition
- Psychology of Music
- Linear Algebra | Machine Learning
- Microprocessor Applications
- Microprocessor System Design | Mobile Robotics | Music History I and II
- Music Informatics | Music Information Retrieval
- Music Theory (3 classes) | Operating Systems
- Physics I and II (with calculus)
- Sound Reinforcement | Statistics
- Switching Circuits

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## ACADEMIC SERVICES

**Conference Reviewing**

- IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)
- International Joint Conference on Neural Networks (IJCNN 2025)
- The International Society for Music Information Retrieval (ISMIR)

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## PUBLICATIONS

- **Faronbi, Daniel**, Iran Roman, and Juan Pablo Bello. "Exploring Approaches to Multi-Task Automatic Synthesizer Programming." *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023.
- **Faronbi, Daniel**; Brown, Phil; 2024; Remote Collaborative Mixing with 3D Audio [PDF]; Music and Audio Research Lab, New York University; Dolby Laboratories; Paper 10708; Available from: https://aes2.org/publications/elibrary-page/?id=22521
- Iran Roman, **Daniel Faronbi**, Isabelle Burger-Weiser, Leila Adu-Gilmore. "F0 ANALYSIS OF GHANAIAN POP SINGING REVEALS PROGRESSIVE ALIGNMENT WITH EQUAL TEMPERAMENT OVER THE PAST THREE DECADES: A CASE STUDY" *SMC 2023 - Sound and Music Computing Conference 2023 (SMC).*
- McFee, B., Matt McVicar, **Daniel Faronbi**, Iran Roman, Matan Gover, Stefan Balke, Scott Seyfarth, Ayoub Malek, Colin Raffel, Vincent Lostanlen, Benjamin van Niekirk, Dana Lee, Frank Cwitkowitz, Frank Zalkow, Oriol Nieto, Dan Ellis, Jack Mason, Kyungyun Lee, Bea Steers, … Waldir Pimenta. (2024). librosa/librosa: 0.10.2.post1 (0.10.2.post1). Zenodo. https://doi.org/10.5281/zenodo.11192913

## DATASETS


- **Daniel Faronbi.** (2023). Multi-Task Automatic Synthesizer Programming Preset Dataset [Data set]. Zenodo. https://doi.org/10.5281/zenodo.7686668